producing P-239 from U-238 requires a good source of slow neutrons in the first place—and the best source of slow neutrons is a controlled nuclear chain reaction. So getting P-239 from U-238 is a lot easier if you already have some U-235.

“turn the United States into one huge factory”: Rhodes 1986, p. 294.

“it would all disappear”: Ibid., p. 275.

“jaundice doesn’t hurt really”: Wigner 1966, interview.

“the rest and the detachment were wonderful”: Ibid.

“‘there will be a chain reaction’”: Ibid.

“Hitler’s success could depend on [nuclear fission]”: Rhodes 1986, p. 281.

Top Policy Group: Ibid., pp. 378 and 387.

“he knew them so well”: Ibid., p. 381.

reason to think so: Daniel Lang 1953, “A Farewell to String and Sealing Wax,” reprinted in *From Hiroshima to the Moon: Chronicles of Life in the Atomic Age*, by Daniel Lang (Simon and Schuster, 1959), p. 58.

“Germany needs me”: David Cassidy 2009, *Beyond Uncertainty: Heisenberg, Quantum Physics, and the Bomb* (Bellevue Literary Press), p. 295.

“machine gun practice in the Bavarian Alps”: Wheeler and Ford 1998, p. 32.

“very casual about numbers”: Jeremy Bernstein 2001, *Hitler’s Uranium Club: The Secret Recordings at Farm Hall*, 2nd ed. (Copernicus), pp. 35–36. Heisenberg’s problems with numbers were well-known among his colleagues. (And, to be clear, Peierls worked with Heisenberg in the 1920s, not on the German bomb program; during the war and afterward, Peierls was in the UK.)

escaped the attention of Heisenberg and his colleagues: Although some people within the German program did seem to understand that purified graphite was a viable option, it is unclear how widely that information was shared, and those who were aware dismissed graphite purification as too expensive to pursue. See ibid., pp. 25–26.

new power source for the Reich’s war engine: Cassidy 2009, p. 322.

never led an experimental team before in his life: Bernstein 2001, p. 40. To be fair, Oppenheimer (the scientific leader of the Manhattan Project) had never been an experimental physicist either—but he had many